(No Model.) 2 Sheets—Sheet 1.

C. G. E. HENNIG.

AIR SHIP.

No. 430,343. Patented June 17, 1890.

Fig: 1.

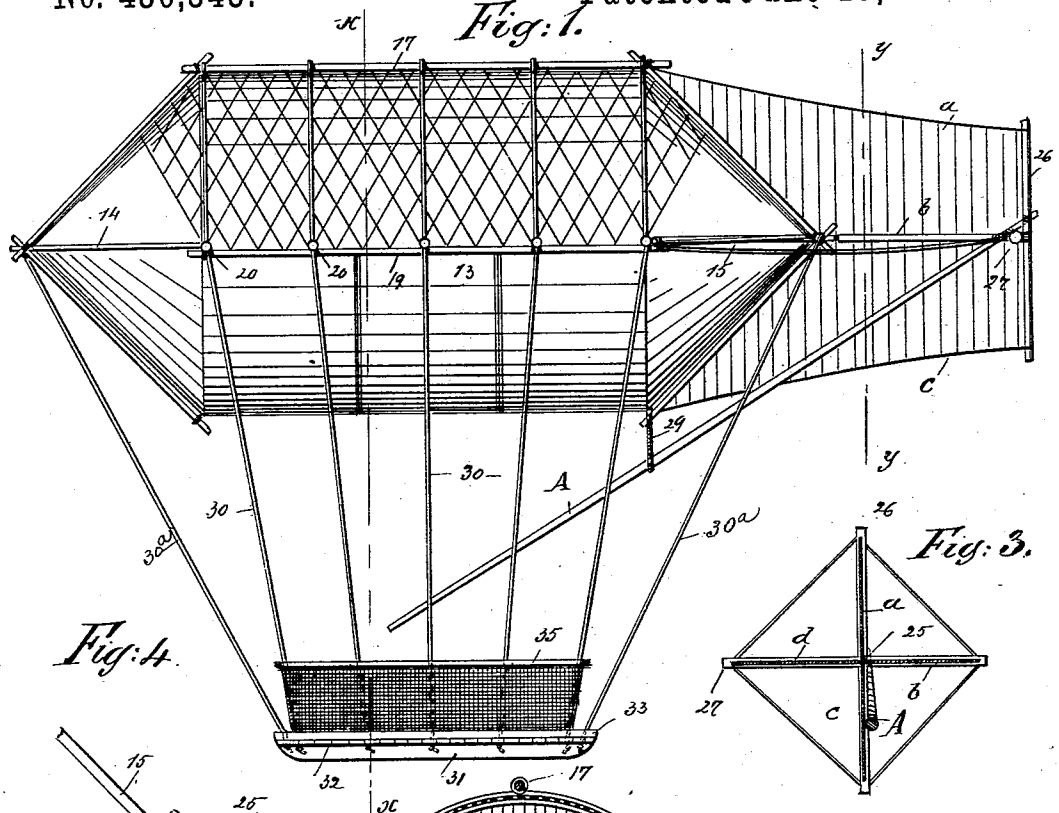
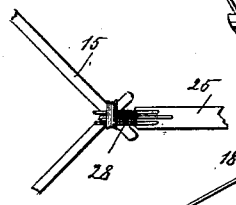


Fig: 3.

Fig: 4.



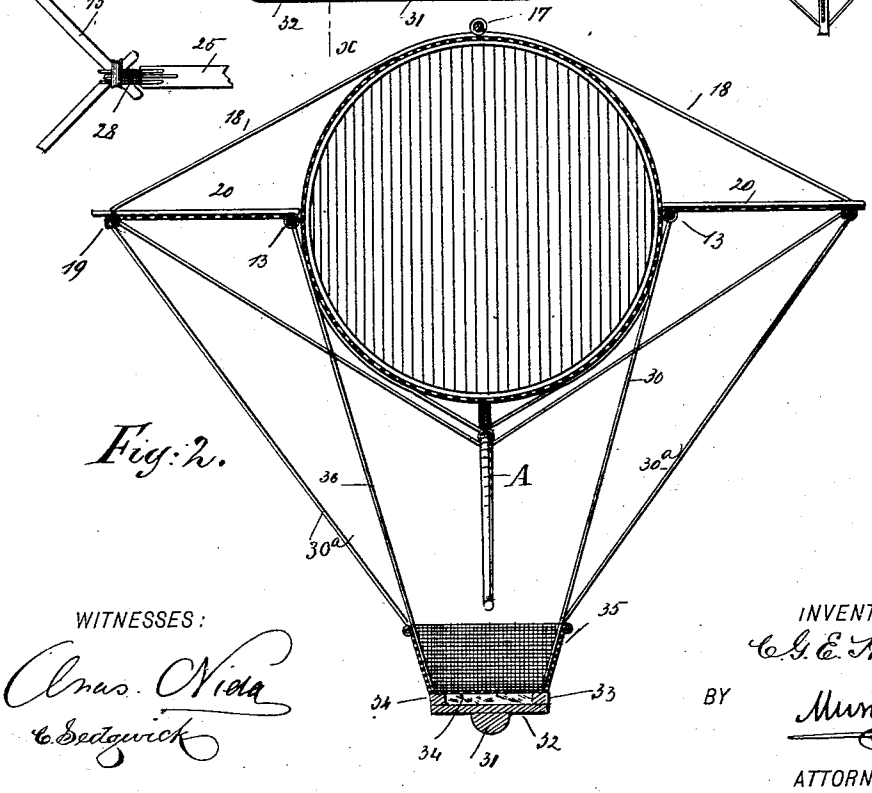


Fig: 2.

WITNESSES:

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C. Sedgwick

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THE NORRIS PETERS CO., PHOTO-LITHO., WASHINGTON, D. C.

(No Model.) 2 Sheets—Sheet 2.

C. G. E. HENNIG.

AIR SHIP.

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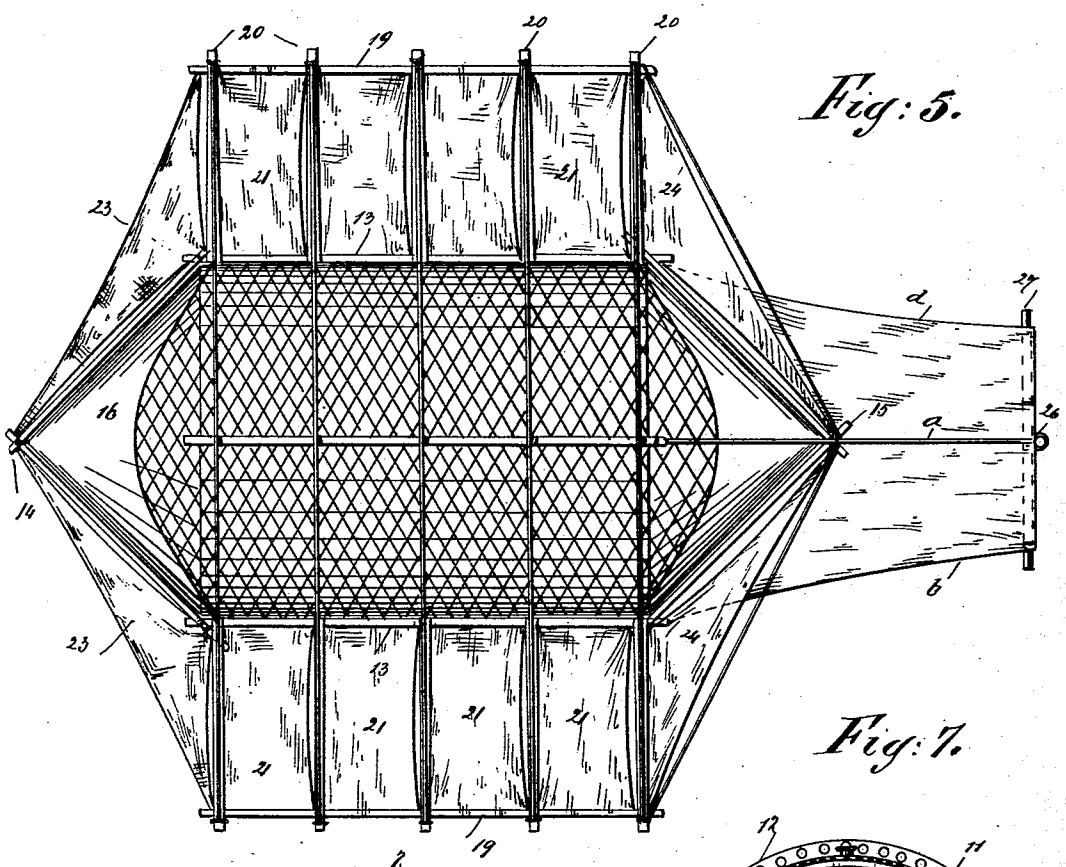

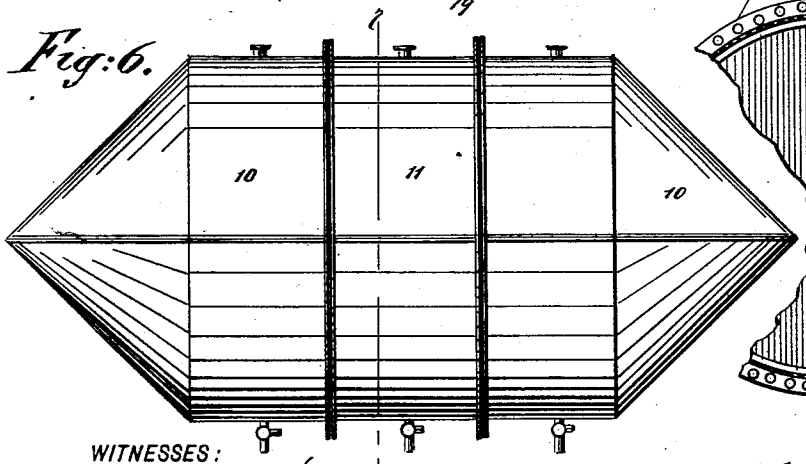

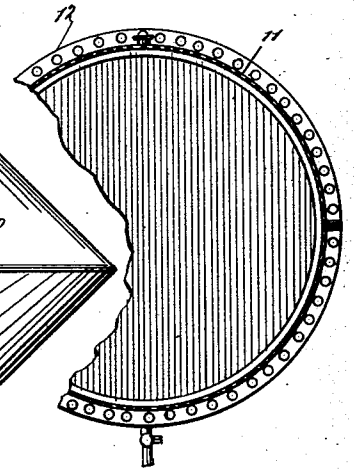

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UNITED STATES PATENT OFFICE.

CARL G. E. HENNIG, OF PATERSON, NEW JERSEY.

AIR-SHIP.

SPECIFICATION forming part of Letters Patent No. 430,343, dated June 17, 1890.

Application filed August 15, 1889. Serial No. 320,806. (No model.)

To all whom it may concern:

Be it known that I, CARL G. E. HENNIG, of Paterson, in the county of Passaic and State of New Jersey, have invented a new and Improved Air-Ship, in which the car, instead of being provided with or rigidly attached to the balloon or gas-receiver, is freely suspended therefrom by ropes or like connections, of which the following is a full, clear, and exact description.

A further object of the invention is to maintain the suspended car in its proper position relatively to the balloon proper or gas-receiver.

The invention consists of certain constructions, arrangements, and combinations of elements to be hereinafter described, and specifically pointed out in the claim.

Reference is to be had to the accompanying drawings, forming a part of this specification, in which similar figures and letters of reference indicate corresponding parts in all the views.

Figure 1 is a side view of my improved air-ship. Fig. 2 is a cross-sectional view looking toward the stern, the view being taken on line *x x* of Fig. 1. Fig. 3 is a cross-sectional view on line *y y* of Fig. 1, the view being given to represent the construction of the rudder. Fig. 4 is an enlarged detail view representing the connection between the rudder and the main frame of the ship. Fig. 5 is a plan view of the ship. Fig. 6 is a detail view of the sectional balloon; and Fig. 7 is a detail sectional view on line *z z* of Fig. 6, representing the eyeleted flange by means of which the balloon-sections are connected.

In constructing an air-ship such as the one forming the subject-matter of this application, I provide a balloon made up of end sections 10, that are preferably conical, and of a center section or sections 11, all the sections being formed with eyeleted flanges 12, through which binding-cords are passed. The several sections are made of silk or of any other proper gas-retaining material.

About the balloon-sections above described I place a frame consisting of two main longitudinal strips 13, the ends of which are united by diagonal bracing-strips 14 and 15. The strips 13 are supported by a net-work 16, and above this net-work I place a backbone or strip 17, to which there are secured guy-ropes 18, that extend diagonally downward to outer poles or strips 19, to which there are bound transverse bars or rods 20, the inner ends of said bars or rods being supported by the strips 13.

To the rods or bars 20 are secured the forward edges of sails 21, the end edges of said sails being secured to the strips 13 and 19, the rear edges being free, so that when the balloon is rising the action of the air upon the sails will tend to carry the ship forward, and the same will be true if the balloon be falling.

To secure additional sail-power, I provide sails 23, which extend from the apex of the frame 14 to the forward ends of the strips 19, and I provide other sails 24, that are secured to the rear bars 20 and to the frame 15, the rear edges of all the sails being free.

In order that the vessel may be guided, I provide a rudder consisting of four sails *a*, *b*, *c*, and *d*, such sails being connected to a short pole 25 and to outer poles 26 and 27, that are arranged at right angles the one to the other and in a plane that is at right angles to the plane occupied by the pole 25. The pole 25 is connected to the frame 15 by a rope joint 28, such joint being formed by passing a series of strands through the pole 25 and about the frame 15.

The tiller A is fulcrumed in a sling 29, and the outer end of this tiller is lashed to the rudder-poles 25 and 26, the tiller-handle extending downward to within reach of the occupants of the car. The car itself is suspended from the frame-work by suspending-ropes 30, and this car is preferably formed with a heavy keel 31, which supports cross-strips 32, carrying longitudinal side strips 33. The space between the side strips is filled in with cork or other buoyant material, as shown at 34, and the sides of the car are made up of canvas or rope netting 35.

In addition to the suspension-ropes 30, I provide guy-ropes 30[a], which extend from each side of the car to the strips 19. Thus the guy-ropes 30[a], in connection with the guy-ropes 18, form an effective means of maintaining the car in proper position relatively to the balloon proper. For instance, should the car swing to one side and the guy-ropes 30[a] on the side to which the car swings thus slacken,

the balloon will immediately rise on that side, and the sail-frames being maintained in position by the guy-ropes 18 and arms 20 the rising of the balloon will immediately take up the slack in the guy-ropes 30[a] and right the car. In other words, by means of my improved arrangement of guy-ropes, in connection with the sail-frames and backbone 17, the sail-frames at each side are effectively maintained in position, as the backbone 17 prevents any strain on the sail-frames of one side from unduly affecting the sail-frame on the opposite side, and any temporary slack in the guy-ropes 30[a] on either side of the car will be taken up by the immediate upward tilting of the balloon proper on that side, the sails on the opposite side acting as the fulcrum until the balloon again exerts an equal strain on the guy-ropes on both sides, when the balloon and car will at once right themselves.

From the construction above described it will be seen that after the balloon has been filled and the ship released the rising of the ship will bring about a pressure upon the sails, and this pressure will act to force the ship forward. After the desired elevation has been reached a portion of the gas is discharged and the ship is permitted to sink, and as the ship so sinks and descends the air, acting upon the sails, will carry it forward, and this operation may be repeated, the ship being forced forward and moving in a zigzag path from a higher to a lower horizonal plane, and vice versa, the forward movement being continually maintained. A proper guidance of the vessel is secured by the manipulation of the rudder, which may be moved so as to guide the vessel to the right or to the left or upward or downward to a certain extent.

Having thus described my invention, I claim as new and desire to secure by Letters Patent—

In an air-ship, the combination, with the balloon having conical ends, a series of horizontal flexible sails around the balloon and free to flex up or down at their rear edges, a car suspended from the balloon, and a rudder formed of four flexible sails *a b c d*, shaped at their inner ends to the contour of the conical rear end of the balloon and secured at the outer corners of their inner ends thereto, of a pole 25, flexibly secured at its inner end to the rear apex of the balloon and extending along the center of the rudder to its outer end, the vertical and transverse poles 26 27 at the rear edges of the said rudder-sails, and the tiller A, secured at the outer end of the rudder-pole 25, flexibly suspended at 29 from the balloon and inclined downwardly therefrom to the car, substantially as set forth.

CARL G. E. HENNIG.

Witnesses:

ALFRED E. FROMMELT,
ALBERT E. HENNIG.